

Course Overview

Course Focus

- Core deep learning architectures & model families
- Applications in control systems, autonomous systems, and generative AI
- Foundations for modern LLMs, reasoning systems, and AI agents

Program Context

- Hands-on labs using world-class tools from NVIDIA, AWS, Intel & IBM
- Preparation for careers in ML, NLP, CV, automation, data science, and robotics

Key Concepts Learned

FOUNDATIONS

- Neural networks, weights, biases, activations, optimization
- Preventing under/overfitting, hyperparameter tuning

CV

- CNNs: LeNet to AlexNet to VGG to ResNet to MobileNet
- Transfer learning for efficient, high-accuracy CV models

SEQUENCE MODELING & NLP

- Tokenization, embeddings, RNNs, LSTMs, GRUs
- Challenges of long-term dependencies; shift to hybrid models

TRANSFORMERS

- Self-attention, multi-head attention, positional encoding
- Scaling laws, zero/few-shot learning, emergence of LLMs

GEN AI MODELS



AUTOENCODERS & VAEs

- Latent space learning, probabilistic representations
- Reconstruction vs generation objectives

GANs

- Generator discriminator adversarial training
- Strengths & challenges (instability)
- GAN applications in synthesis, translation and segmentation

DIFFUSION MODELS

- Noise to image denoising generation
- Stable training and high-quality output
- Latent diffusion & text-conditioned synthesis

RL & AI AGENTS



REINFORCEMENT LEARNING BASIS

- States, actions, rewards, value functions
- Exploration & exploitation

MODERN AI AGENTS

- Reason + Action is ReAct
- Chain-of-Thought
- Agents search, APIs, databases, documents
- Memory systems: short-term, episode, semantic, procedure

REASONING & RETRIEVAL RAG

- Embeddings, similarity search, vector databases
- Using external knowledge to reduce hallucinations
- Integrating RAG with agency

Final Project

Key Components

- Specialized agents: research, critic, summarizer, safety
- Reasoning techniques: CoT, ReAct, Reflection, RAG-based grounding
- Tool integration: search, OCR, document processing, external API
- Learning elements: feedback loops & reward-style evaluation
- Safety: input validation, boundary enforcement, transparent reasoning

Multi-Agent Health Summary System

Goal: Help aging adults understand medical records, symptoms, and doctor’s guidance using AI-supported health summaries and overall pictures.

